Differentiation Rules: Power, Product, Quotient, Chain

Section 1: Power Rule and Basic Operations

1 Find the derivative of $f(x) = 4x^5 - 3x^3 + 7x - 9$.

a. $20x^4 - 9x^2 + 7$

b. $20x^4 - 9x^2$

c. $4x^4 - 3x^2 + 7$

d. $20x^4 - 6x^2 + 7$

2 Differentiate $g(x) = \frac{4}{\sqrt{x}} + \frac{3}{x^2}$.

a. $-2x^{-3/2} - 6x^{-3}$

b. $-2x^{-1/2} - 6x^{-3}$

c. $2x^{-3/2} - 6x^{-3}$

d. $-4x^{-3/2} - 3x^{-3}$

3 Find all points where the curve $y = x^3 - 3x^2 - 9x + 5$ has a horizontal tangent.

a. $x = -1$ and $x = 3$

b. $x = 1$ and $x = -3$

c. $x = 0$ and $x = 3$

d. $x = -1$ only

Section 2: Product and Quotient Rules

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4 Differentiate $h(x) = (x^2 + 3)(2x - 5)$.

a. $6x^2 - 10x + 6$

b. $6x^2 - 10x$

c. $4x^2 - 10x + 6$

d. $6x^2 + 6$

5 Find $\frac{dy}{dx}$ for $y = \frac{3x-1}{2x+5}$.

a. $\frac{17}{(2x+5)^2}$

b. $\frac{1}{(2x+5)^2}$

c. $\frac{6x+17}{(2x+5)^2}$

d. $\frac{3}{2}$

6 If $f(2) = 3, f'(2) = -1, g(2) = 4, g'(2) = 2$, find $\left(\frac{f}{g}\right)'(2)$.

a. $-10/16 = -5/8$

b. $2/16 = 1/8$

c. $-2/16$

d. $-1/2$

Section 3: The Chain Rule

7 Find $\frac{d}{dx}[(3x^2 - 5x + 2)^4]$.

a. $4(3x^2 - 5x + 2)^3(6x - 5)$

b. $4(3x^2 - 5x + 2)^3$

c. $12(6x - 5)^3$

d. $4(6x - 5)^3$

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8 Find $\frac{dy}{dx}$ for $y = \sqrt{4 - x^2}$.

a. $\frac{-x}{\sqrt{4 - x^2}}$

b. $\frac{x}{\sqrt{4 - x^2}}$

c. $\frac{-1}{2\sqrt{4 - x^2}}$

d. $\frac{-2x}{\sqrt{4 - x^2}}$

9 If $h(x) = f(g(x))$ with $g(3) = 5$, $g'(3) = 4$, and $f'(5) = -2$, find $h'(3)$.

a. -8

b. 8

c. -10

d. 20

Section 4: Trigonometric Derivatives with Chain Rule

10 Differentiate $y = \sin(3x^2)$.

a. $6x\cos(3x^2)$

b. $3\cos(3x^2)$

c. $\cos(6x)$

d. $6x\sin(3x^2)$

11 Find $\frac{d}{dx}[\tan^3(x)]$.

a. $3\tan^2(x)\sec^2(x)$

b. $3\tan^2(x)$

c. $3\sec^4(x)$

d. $\sec^6(x)$

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12 Find the second derivative y'' for $y = \cos(2x)$.

a. $-4\cos(2x)$

b. $4\cos(2x)$

c. $-2\sin(2x)$

d. $-4\sin(2x)$

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